**ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES**
**(Autonomous)**
Thallapaka Panchayath, New Boyanapalli, Rajampet, Annamayya Dist., A.P. - 516126
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**DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (ARTIFICIAL INTELLIGENCE)**

**Academic Year: 2025-2026**

**Batch No: 2**

**PROJECT TITLE**: "Enhancing Emergency Care Through Machine Learning-Based Admission Prioritization"

**ABSTRACT**: The use of artificial intelligence (AI) and machine learning (ML) in emergency medicine has grown rapidly in recent years. Studies in this field can be grouped into three main areas: prediction and detection of disease, prediction of hospital admission, discharge, and mortality, and the development of machine learning–based triage systems. In the first category, various algorithms such as logistic regression, Naïve Bayes, random forests, multilayer perceptrons (MLP), support vector classifiers (SVC), and long short-term memory networks (LSTM) have been applied to datasets derived from electronic health records (EHRs) and medical imaging. The second category centers on predicting patient outcomes, including admission, discharge, and mortality, using data from EHRs, laboratory tests, and vital signs, with algorithms such as decision trees, random forests, and logistic regression models being commonly employed. The third category involves the creation of machine learning–based triage systems for emergency departments, utilizing patient data such as vital signs and medical history, with approaches ranging from decision trees and artificial neural networks to fuzzy logic models. Collectively, these studies demonstrate the significant potential of AI and ML in enhancing emergency medicine, though the accuracy and effectiveness of such algorithms remain highly dependent on the quality and volume of available data. Further research is needed to validate existing findings and improve algorithmic performance in clinical practice.

**Keywords:** Logistic Regression, Naive Bias, Random Forest, MLP, SVC, LSTM.

## SYSTEM SPECIFICATIONS:

### H/W Specifications:

| | |
|---|---|
| **Processor** | I5/Intel Processor |
| **RAM** | 8GB |
| **Hard Disk** | 128 GB |

**S/W Specifications:**

| | |
|---|---|
| **Operating System** | Windows 10/11 |
| **Server-side Script** | Python (Flask/FastAPI) |
| **Languages** | Python |
| **IDE** | Python Jupyter, VS Code, |
| **Libraries Used** | Tensorflow, keras, pandas, scikit-learn, numpy, matplotlib, imbalanced-learn. |

## Project Associates:

| Roll Number | Name | Signature |
|---|---|---|
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**PROJECT GUIDE**
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